

Menyederhanakan Surds (Memanfaatkan Faktor & Kelipatan)

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Topic: Menyederhanakan Surds (Memanfaatkan Faktor & Kelipatan)

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Lesson Plan

Learning Objectives

- Deconstruct surds into prime factor products to identify square components.
- Apply the product property of radicals, $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$, to simplify complex expressions.
- Evaluate numerical values to instantly identify the largest perfect square factor (4, 9, 16, 25, etc.).
- Simplify non-perfect square radicals into their simplest mixed radical form.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Bintang! Great to see you. Before we dive into today's challenge, let's sharpen our tools with a quick 4-question sprint. We're looking at our previous visual surds and our business-scale place values. Ready? Question 1: Visual check—if a square has an area of 50, between which two whole numbers does its side length lie? Question 2: Simplify the visual—draw a square of area 18; how many ' $\sqrt{2}$ ' units fit in its side? Question 3: Business scale—if a company revenue is 1.2 billion, how many 'millions' are inside that? Question 4: Place value—write 45 million in scientific notation. Type your answers in the chat!

Activities:

- *Rapid-fire chat box answers*
- *Mental estimation of square roots*
- *Place value conversion for large business figures*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Bintang, imagine you are a data analyst for a global tech firm. You are calculating the 'Standard Deviation' of market volatility. The formula gives you an answer like $\sqrt{200}$. Reporting ' $\sqrt{200}$ ' to your CEO looks messy. We need to simplify it to ' $10\sqrt{2}$ ' to make it comparable to other data sets. Today, we turn messy square roots into clean, professional 'Mixed Surds'.

Activities:

- Discussion on why simplified forms are preferred in professional fields

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

To do this, we need to be 'Factor Detectives'. Tell me, Bintang, what are the first 6 perfect squares? (Wait for: 1, 4, 9, 16, 25, 36). Perfect. Remember, a square root 'neutralizes' a square. So, $\sqrt{25}$ is just 5. We will use this 'neutralizing power' to pull numbers out of the radical sign.

Activities:

- Reciting perfect squares
- Mental math factor pairs

4. Problem Introduction (10 minutes)

Teacher Script:

Let's look at $\sqrt{12}$. It isn't a perfect square. But, we can use the 'Breakup Rule': $\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$. I can write 12 as $4 \cdot 3$. So, $\sqrt{12} = \sqrt{4} \cdot \sqrt{3}$. Since we know $\sqrt{4}$ is 2, the final answer is $2\sqrt{3}$. Why didn't I use $6 \cdot 2$? Because 6 and 2 aren't square numbers! They don't help us simplify. We always hunt for the 'Hidden Square'.

Activities:

- Demonstrating the product property of radicals
- Comparing factor pairs (4×3 vs 6×2)

5. Guided Practice (15 minutes)

Teacher Script:

Let's try a harder one together: $\sqrt{72}$. On the whiteboard, list all factor pairs of 72. (9×8 , 36×2 , 4×18). Which is the 'King' of these pairs? The one with the biggest perfect square! Yes, 36. Now, apply the rule: $\sqrt{36 \cdot 2}$. Break it apart. What do you get? (Wait for: $6\sqrt{2}$). Excellent. Now, what if we used 9×8 ? We would get $3\sqrt{8}$, but $\sqrt{8}$ can be simplified again! It's always faster to find the biggest square first.

Activities:

- Factor tree construction
- Multi-step simplification practice

6. Independent Practice (15 minutes)

Teacher Script:

Now, it's your turn to be the lead analyst. I'm putting 5 'messy' market figures on the board: $\sqrt{20}$, $\sqrt{45}$, $\sqrt{48}$, $\sqrt{75}$, and the boss-level $\sqrt{108}$. Simplify these as fast as you can. Type your final forms in the chat or write them on the board.

Activities:

- *Solo problem solving*
- *Timed challenge for speed (fluency building)*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Incredible work today, Bintang. You've mastered the art of extracting squares from surds. Remember: factor, split, and simplify. Before we go, please log in to <https://learn.algonova.id/login> to access your post-session notes and homework. What was the most 'aha!' moment for you today?

Activities:

- *Verbal summary*
- *Platform login reminder*

Homework

Medium Questions:

1. Simplify $\sqrt{50}$
2. Simplify $\sqrt{28}$
3. Simplify $\sqrt{63}$
4. Simplify $\sqrt{80}$

Hard Questions:

1. Simplify $\sqrt{162}$
2. Simplify $\sqrt{300}$
3. Simplify $3 * \sqrt{48}$ (Hint: simplify the root then multiply by 3)
4. Simplify $\sqrt{98} + \sqrt{18}$
5. Find the value of x if $\sqrt{x} = 5\sqrt{2}$
6. Simplify $\sqrt{252}$ using prime factorization

Answer Key:

$$\sqrt{50} = 5\sqrt{2}$$
$$\sqrt{28} = 2\sqrt{7}$$

$$\sqrt{63} = 3\sqrt{7}$$

$$\sqrt{80} = 4\sqrt{5}$$

$$\sqrt{162} = 9\sqrt{2}$$

$$\sqrt{300} = 10\sqrt{3}$$

$$3 * \sqrt{48} = 12\sqrt{3}$$

$$\sqrt{98} + \sqrt{18} = 7\sqrt{2} + 3\sqrt{2} = 10\sqrt{2}$$

$$\sqrt{x} = \sqrt{25*2} \rightarrow x = 50$$

$$\sqrt{252} = \sqrt{36 * 7} = 6\sqrt{7}$$

REVIEW: Membongkar 'Surds' Secara Visual - $\sqrt{40}$ is between 6 and 7 (Answer: $6 < \sqrt{40} < 7$)

REVIEW: Membongkar 'Surds' Secara Visual - Side of area 20 square = $2\sqrt{5}$

REVIEW: Nilai Tempat & Bilangan Besar - 15.5 Million in Standard Form = 15,500,000

REVIEW: Nilai Tempat & Bilangan Besar - \$4.2 Billion / 100 = 42 Million

Parent Reminder

Bintang's Progress: Mastering Surds

Today Bintang learned how to simplify complex square roots (surds) by breaking them down into factors. This is a crucial skill for Grade 10 Math and beyond. He practiced identifying 'hidden' square numbers to make expressions cleaner and easier to use in business calculations. Please ensure he completes the homework on the portal to reinforce these speed-skills!